

like `somelist[-1]` and `somelist[-2]`, which access the last and second-to-last values in a list, respectively. This is useful for finding the disk currently at the top of the tower.

```
1. """The Tower of Hanoi, by Al Sweigart al@inventwithpython.com
2. A stack-moving puzzle game.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: short, game, puzzle"""
5.
6. import copy
7. import sys
8.
9. TOTAL_DISKS = 5 # More disks means a more difficult puzzle.
10.
11. # Start with all disks on tower A:
12. COMPLETE_TOWER = list(range(TOTAL_DISKS, 0, -1))
13.
14.
15. def main():
16.     print("""The Tower of Hanoi, by Al Sweigart al@inventwithpython.com
17.
18. Move the tower of disks, one disk at a time, to another tower. Larger
19. disks cannot rest on top of a smaller disk.
20.
21. More info at https://en.wikipedia.org/wiki/Tower_of_Hanoi
22. """)
23.
24.
25.     # Set up the towers. The end of the list is the top of the tower.
26.     towers = {'A': copy.copy(COMPLETE_TOWER), 'B': [], 'C': []}
27.
28.     while True: # Run a single turn.
29.         # Display the towers and disks:
30.         displayTowers(towers)
31.
32.         # Ask the user for a move:
33.         fromTower, toTower = askForPlayerMove(towers)
34.
35.         # Move the top disk from fromTower to toTower:
36.         disk = towers[fromTower].pop()
37.         towers[toTower].append(disk)
38.
39.         # Check if the user has solved the puzzle:
40.         if COMPLETE_TOWER in (towers['B'], towers['C']):
41.             displayTowers(towers) # Display the towers one last time.
42.             print('You have solved the puzzle! Well done!')
43.             sys.exit()
44.
45.
46. def askForPlayerMove(towers):
47.     """Asks the player for a move. Returns (fromTower, toTower)."""
48.
49.     while True: # Keep asking player until they enter a valid move.
50.         print('Enter the letters of "from" and "to" towers, or QUIT.')
```